

✓ Control Structures R

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```
for(i in 1:5) {print(i^2)}
```

```
[1] 1  
[1] 4  
[1] 9  
[1] 16  
[1] 25
```

```
for(i in c(2,4,6,7)) {print(i^2)}
```

```
[1] 4  
[1] 16  
[1] 36  
[1] 49
```

```
x=c(2,4,6,8,10,12)  
excount=function(x){  
  count=0  
  for(xval in x){  
    if(xval/2>3)  
      count=count+1  
  }  
  print(count)  
}  
excount(x)
```

```
[1] 3
```

```
child = c("child1", "child2", "child3")  
sweet = c("sweet1", "sweet2", "sweet3")  
for (x in child) {  
  for (y in sweet) {  
    print(paste(x, y))  
  }  
}
```

```
[1] "child1 sweet1"  
[1] "child1 sweet2"  
[1] "child1 sweet3"  
[1] "child2 sweet1"  
[1] "child2 sweet2"  
[1] "child2 sweet3"
```

```
[1] "child3 sweet1"
[1] "child3 sweet2"
[1] "child3 sweet3"
```

```
for(x in child){
  for(y in sweet){
    print(paste(x,y))
  }
}
```

```
[1] "child1 sweet1"
[1] "child1 sweet2"
[1] "child1 sweet3"
[1] "child2 sweet1"
[1] "child2 sweet2"
[1] "child2 sweet3"
[1] "child3 sweet1"
[1] "child3 sweet2"
[1] "child3 sweet3"
```

```
drink = c("coffee", "lemonade", "tea", "juice")
```

```
for(x in drink){
  if(x=="tea"){
    break
  }
  print(x)
}
```

```
[1] "coffee"
[1] "lemonade"
```

```
for(x in drink){
  if(x=="lemonade"){
    next
  }
  print(x)
}
```

```
[1] "coffee"
[1] "tea"
[1] "juice"
```

```
i=1
while(i<10){
  print(i^2)
  i=i+2
}
```

```
[1] 1
[1] 9
```

```
[1] 25
[1] 49
[1] 81
```

```
sumfunction = function(){
  sum = 0
  number = as.integer(readline(prompt="Please
select any number less than 25: "))
  while (number <= 25) {
    sum = sum + number
    number = number + 1 }
  print(paste("The sum of numbers received from
the While Loop: ", sum))
}
sumfunction()
```

```
Please
select any number less than 25: 22
[1] "The sum of numbers received from \nthe While Loop: 94"
```

```
i= 1
repeat{
  print( i^2 )
  i = i+2
  if ( i > 10 )
    break
}
```

```
[1] 1
[1] 9
[1] 25
[1] 49
[1] 81
```

```
i = 1
repeat{
  i = i+1
  if (i < 10) next
  print(i^2)
  if (i >= 13) break
}
```

```
[1] 100
[1] 121
[1] 144
[1] 169
```

```
abc=function(x){
  x^2
```

```
}  
abc(3)
```

9

```
abc=function(x,y){  
  x^2+y^2  
}  
abc(3,4)
```

25

```
abc = function(x){  
  sin(x)^2+cos(x)^2 + x  
}  
abc(9)
```

10

```
abc = function() {  
  for(i in 1:3) {  
    print(i^3)  
  }  
}  
abc()
```

```
[1] 1  
[1] 8  
[1] 27
```

```
seq(from=2,to=4)
```

2 · 3 · 4

```
seq(from=4,to=2)
```

4 · 3 · 2

```
seq(from=-4,to=4)
```

-4 · -3 · -2 · -1 · 0 · 1 · 2 · 3 · 4

```
seq(from=10,to=20,by=2)
```

10 · 12 · 14 · 16 · 18 · 20

```
seq(from=20,to=10,by=-2)
```

```
20 · 18 · 16 · 14 · 12 · 10
```

```
seq(from=3, to=-2, by=-0.5)
```

```
3 · 2.5 · 2 · 1.5 · 1 · 0.5 · 0 · -0.5 · -1 · -1.5 · -2
```

```
seq(to=10, length=10)
```

```
1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10
```

```
seq(from=10, length=10)
```

```
10 · 11 · 12 · 13 · 14 · 15 · 16 · 17 · 18 · 19
```

```
seq(from=10, length=10, by=0.1)
```

```
10 · 10.1 · 10.2 · 10.3 · 10.4 · 10.5 · 10.6 · 10.7 · 10.8 · 10.9
```

```
seq(from=10, length=10, by=-2)
```

```
10 · 8 · 6 · 4 · 2 · 0 · -2 · -4 · -6 · -8
```

```
seq(from=10, length=5, by=-.2)
```

```
10 · 9.8 · 9.6 · 9.4 · 9.2
```